

ANAKIN DEY

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EDUCATION

University of Illinois Urbana-Champaign

Bachelors of Science in Mathematics, Minor in Computer Science

May 2024

GPA: 3.93/4.00

EXPERIENCE

Undergraduate Research Assistant: *Texas State University Math REU*

Jun 2023 – Present

- Researching finite group classification using both mathematical and computational tools with [Prof. Yong Yang](#)
- Designing, proving, and implementing group theoretic algorithms using GAP and SageMath which has yielded brand new classifications of fundamental structures in mathematics
- Using computer algebra systems and cluster computing to automate and manage the search for these structures which resulted in a more efficient cycle of improving and checking results

Software Engineering Intern: *CME Group*

Jun 2022 – Aug 2022

- Refactored large code bases in order to support multiple file formats and platforms for tracing and logging of messages, resulting in tracing having 20% less impact on the performance of applications
- Used Jenkins, XL Release, and in-house tools to benchmark the performance impact from using the new formats
- Explored the use of dashboards and parsers to better process traces and give new insights on system performance

Undergraduate Research Assistant: [Learning, Decision, Control, Autonomy Lab](#)

Aug 2021 – Present

- Researching algorithms and heuristics for multi-target and multi-agent planning and movement under multiple constraints as well as deceptive and adversarial movement over graphs with [Prof. Melkior Ornik](#)
- Combining concepts from heuristic approximation algorithms, Markov decision processes, probability theory, geometry, and graph theory to design new and efficient heuristics to use with various algorithms
- Benchmarking algorithms in Python and using Matplotlib, NetworkX, and OSMnx to create diagrams and charts
- Maintaining high code quality with various tools and workflows such as Black, pytest, Flake8, Coverage.py, and mypy

Software Engineering Intern: *SiteIQ*

May 2021 – Aug 2021

- Improved the accuracy of status reports from multiple fueling stations using Python and JavaScript by streaming and parsing data sources such as two-wire serial and ZMODEM files
- Identified error codes and data that were not described in given documentation but present in 50% of devices and logged them for further analysis by the team

Course Assistant: *UIUC*

Jan 2021 – Present

- Course Assistant for Introduction to Computer Science (Jan 2021 - May 2021), Discrete Structures (Aug 2021 – Present), Introduction to Algorithms (Jan 2022 – Dec 2022) and Models of Computation (Jan 2023 - Present)
- Teaching theoretical computer science concepts such as state machines, divide-and-conquer, dynamic programming, and graph algorithms by presenting topics in a clear and concise manner

PUBLICATIONS

Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency

Anakin Dey, Melkior Ornik ([IEEE ISC2 2022](#))

- Designed a new multi-agent heuristic algorithm to solve a weighted path planning problem in the context of outage restoration after natural disasters which was up to 50% better than existing heuristics
- Paper on [arXiv](#) and source code on [GitHub](#)

EXTRACURRICULARS

Founder & President: SIGma | **Special Interest Group for Math and Algorithms** | cstheory.org

- Founded club to teach theoretical computer science and mathematics to students of various backgrounds
- Creating weekly meetings covering ideas such as efficient [binary generation](#), [decidability](#) and Turing Machines, [state machines](#), and other [mathematical](#) concepts in a comprehensive and intuitive manner

Admin: SIGPwny | **Special Interest Group for Security** | sigpwny.com

- Hosted [2 meetings](#) on cryptographical concepts including block ciphers, RSA, Diffie-Hellman, and elliptic-curve cryptography, complete with multiple challenges, to over 70 people per meeting
- Created cryptography challenges and reverse engineering for [UIUCTF 2021](#) / [2022](#) / [2023](#) for over 1400 participants and [Fall CTF 2021](#) / [2022](#) for over 300 participants

TECHNICAL SKILLS

Languages: Python + SageMath, GAP, Java, C++, JavaScript, TypeScript, Agda, C

Tools / Frameworks: Bash, Git, JUnit, mypy, pytest, Matplotlib, NetworkX, Numpy, GDB, Catch2, Node.js, Pandas, Jira, Grafana